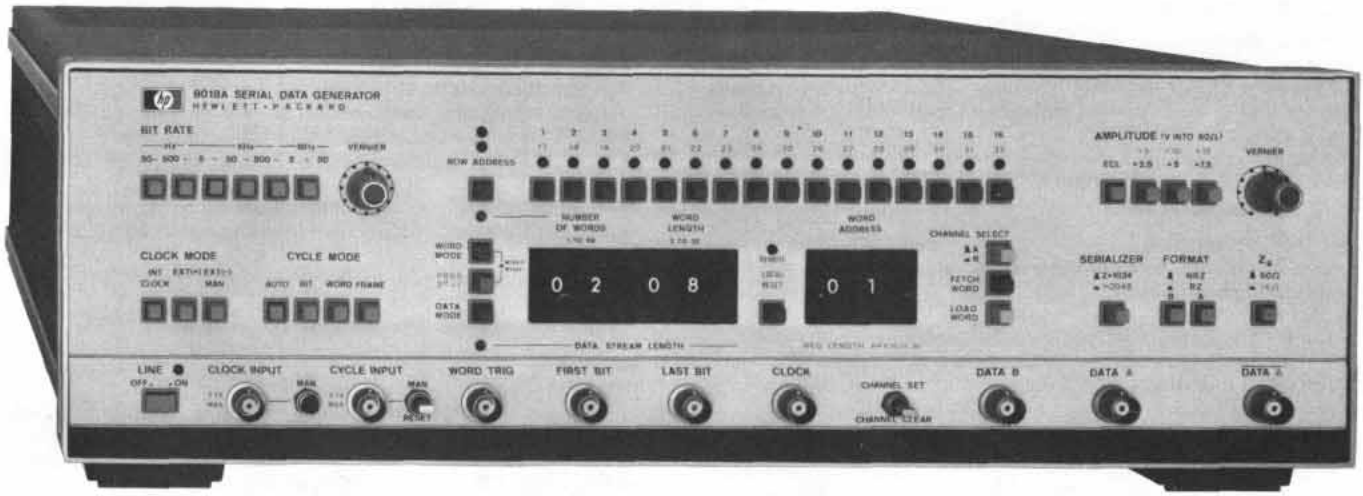
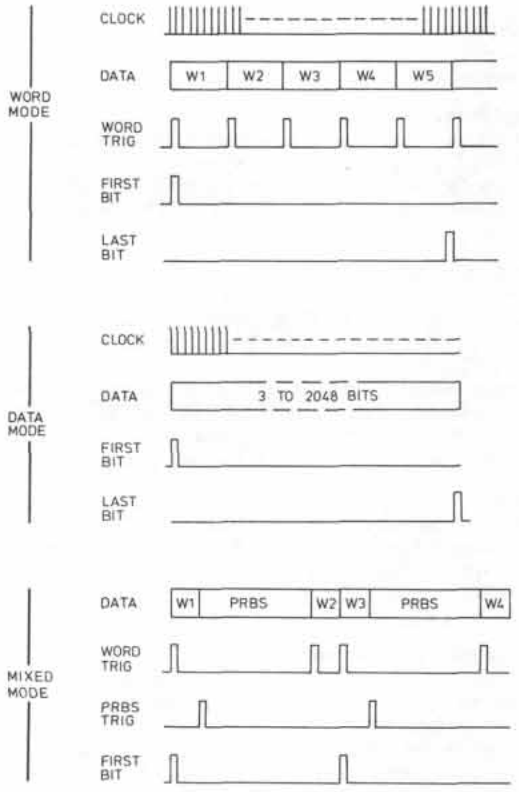
WORD GENERATORS

50 MHz serial data/PRBS generator

Model 8018A

- 2048 bit, dual channel memory
- Variable word and pattern length
- TTL, ECL, CMOS compatible
- PRBS generation
- HP-IB interface
- Mixed PRBS/word output for serial data links





The Hewlett-Packard model 8018A is a high performance data generator designed to meet all of your requirements for serial stimulus up to 50 Mbits/s. Its dual channel memory, for example, contains 2048 individually programmable serial bits, sufficient capacity for the most complex data pattern requirements. Both word and data stream length are variable so you can configure data streams that exactly match your testing applications.

Pattern generating capability is enhanced by a Pseudo-Random Binary Sequency (PRBS) generator. PRBS is a convenient means of generating "worst-case" test patterns and extends pattern length to over 1 million bits. An innovative new technique even lets you mix PRBS and programmable data words in a single stream, perfect for simulating preamble-data-postamble patterns in serial data link applications.

A high performance output amplifier adds to the 8018A's wide applicability. It delivers clean, 6 ns pulses with repetition rates from dc to 50 Mbits/s. Output amplitude is variable up to 15 volts into 50Ω. This enables you to directly drive logic circuits ranging from TTL to CMOS. Output levels for emitter-coupled-logic (ECL) are also provided.

This wide range of operating modes has been designed to shorten and simplify troubleshooting tasks whenever a serial data source is required. In aerospace, telecommunications, integrated circuits, and in computer and peripheral interfacing, the 8018A provides the stimulus you need for digital design and testing.

For production and other systems environments an HP-IB programming interface (option 001) provides remote control of data generating functions. The interface controller can be anything from a large computer system to a simple card reader. The 15263A card reader (option 002) is especially suited to this purpose. Cards can be marked with instructions or data for rapid and error free memory loading. A 4-channel adaptor (HP15450A) and 4-channel TTL-CMOS translator (15451A) are also available as accessories for the 8018A (see page 343).

Specifications

Word and Data Generation

Number of channels: 2.
Channel length: 1024 bits (2048 bit total memory capacity).
Word length (M): 3 to 32 bits (to 2048 bits in Data mode).
Number of words (N): variable from 1 to 99.
Channel serialization: channels may be cascaded to extend Channel A length to 2048 bits.
Data content: each bit is individually programmable using front panel switches, or remotely via optional HP-IB interface.
Data formatting: RZ and NRZ formats independently selectable for both output channels. Width in RZ format approximates width of clock output pulse.
Data generation modes
Word mode: data frame consists of N words of length M bits/word.
Data mode: data frame consists of a continuous pattern of length between 3 and 2048 bits. Frame length is determined by 4-digit number set into thumbwheel switches.

PRBS mode: Pseudo-Random Binary Sequence of length 2^n-1 bits is produced: n = 9, 10, 15, 20.
Mixed mode: Same as WORD mode except with PRBS sequence inserted after every odd number word. Simulates preamble, data message, postamble.
Frame length: 1 to 99 words (WORD mode) or 3 to 2048 bits (DATA mode).
Channel set/clear: fills selected data channel with ones or zeros.

Data Outputs

DATA A, $\overline{\text{DATA A}}$

Output Attenuator Positions	RS	RL	Maximum Amplitude	Minimum Amplitude	tr/tf (10%-90%)	Maximum Repetition Rate
2.5 V	50Ω	50Ω	7.5 V	1.3 V	6 ns	50 MHz
5 V	1 KΩ	50Ω	15 V	2.5 V	8 ns	40 MHz
7.5 V	50Ω	1 KΩ	15 V	2.5 V	8 ns	40 MHz
ECL	50Ω	50Ω	1.0 V	0.5 V	5 ns	50 MHz

Pulse amplitude: variable in three ranges from 1.3 V to 15 V plus fixed ECL position. See table.
Output format: simultaneous Data and $\overline{\text{Data}}$ waveforms are provided. Data output is positive-going with OV baseline. $\overline{\text{Data}}$ is inverted with identical upper and lower level voltages.
ECL position: positive-going pulse with 0.6 to 1.0 V amplitude, and +0.5 to −1.6 V offset. Amplitude and offset internally adjustable. 5 ns maximum transition time. Levels preset for standard ECL. (50Ω source and load resistance).
Maximum transition times (10-90%): 6 ns. See table.
Maximum preshoot, overshoot, pulse top/baseline distortion: 10% of amplitude. 15% in ECL position.
Source resistance: selectable 50Ω or 1 KΩ.
Relation to clock pulse: leading edge of Channel B output coincides with leading edge of clock output ±3 ns.
Overload protection: cannot be damaged by externally applied voltages between 0 and 16 volts. Protected against open and short circuits.

DATA B

Pulse amplitude: 2.4 V min. into 50Ω, 4.8 V min. into open circuit.
Polarity: positive.
Source resistance: 50Ω.
Relation to clock pulse: leading edge of Channel B output coincides with leading edge of clock output ± 3 ns.
Overload protection: cannot be damaged by externally applied voltages between +5 and −2 volts. Additionally protected against voltages between 0 and 16 volts when current limited to 20 mA. Protected against open circuit and shorts to ground.

Synchronizing Outputs

Clock: RZ pulse, occurs with each data bit.
First bit: RZ pulse, identifies first bit of data pattern.
Last bit: RZ pulse, identifies last bit of data pattern.
Word trigger: RZ pulse, identifies first bit of each word.
PRBS trigger: NRZ pulse, identifies beginning of each PRBS pattern.
Amplitude
Clock: 2.4 V min. into 50Ω, 4.8 V min. into open circuit.
FB, LB, WT, PRBS TRIG: 1.2 V min. into 50Ω, 2.4 V min. into open circuit.
Source resistance: 50Ω.
Width:
Clock, FB, LB, WT: 50% ± 20% of period in internal clock mode. Approximates width of externally applied clock pulse in external clock mode.
PRBS trigger: 3 clock cycles.
Overload protection: cannot be damaged by externally applied voltages between +5 and −2 volts. Additionally protected against voltages between 0 and 16 volts when current limited to 20 mA. Protected against open circuits and shorts to ground.

Clocking

Internal
Bit rate: 50 Hz to 50 MHz (40 MHz max. in MIXED mode).
Jitter: 0.2% + 50 ps.
Controls: 5 ranges and 3 turn potentiometer for fine adjust.

External clock input
Bit rate: DC to 50 MHz (40 MHz max. in MIXED mode).
Nominal trigger level: 0.5 V (EXT+), −1.2 V (EXT−).
Minimum pulse amplitude: 1.0 V (EXT+), 0.8 V (EXT−).
Trigger slope: positive.
Minimum pulse width: 10 ns.
Input resistance: 50Ω to ground.
Overload protection: ±7 V. 0-16 V when current limited to 20 mA. By means of an internal switch, the CLOCK input may be switched to a high impedance mode. The following specifications then apply.
Input resistance: 1 LS-TTL load in series with 300Ω.
Bit rate: DC to 40 MHz.
Trigger pulse: TTL levels. Amplitude may be increased to 16 volts when current limited to 20 mA.
Minimum pulse width: 15 ns.
Manual: pushbutton switch enables single bit output.

Cycle Modes

Auto: data frame recycles continuously.
Bit: single bits are triggered by pulses at the CYCLE INPUT. If the input is held high, data bits are continuously generated. Data generation ceases when the input goes low and continues from where it stopped when the input is returned to the high state.
Word: single words are triggered by pulses at the CYCLE INPUT. If the input is held high, words are continuously generated. When the input goes low, data generation ceases after completion of the current word. Data generation continues with the next word when the CYCLE INPUT is returned to the high state.
Frame: single data frames are generated by pulses at the cycle input. If the input is held high, frames are continuously generated. When the input goes low, the current frame is completed and data generation ceases.
Cycle input
Nominal trigger level: 0.5 V.
Trigger slope: positive.
Minimum pulse amplitude: 1.0 V.
Minimum pulse width: 10 ns.
Input resistance: 50Ω to ground.
Overload protection: ±7 V. 0-16 V when current limited to 20 mA. By means of an internal switch, the CYCLE INPUT can be set to high impedance. The following specifications apply.
Input impedance: 1 LS-TTL load in series with 300Ω.
Trigger pulse: TTL levels.
Minimum pulse width: 15 ns.
Manual: Switch enables outputting single bits, words, or frames.
Reset: Returns generator to bit1.

General

Power requirements: 100 V, 120 V, 220 V, or 240 V; ±5 to −10%, 48 to 440 Hz. 230 V A max.
Environmental: 0 to 50°C, and with relative humidity to 95% at 40°C.
Weight: net 12 kg (26.5 lbs). Shipping 16 kg (35.3 lbs).
Size: 133 H x 426 W × 422 mm D (5.2″ × 16.8″ × 16.6″).

Options and Accessories

	Price
Opt 001: HP-IB Interface. Permits loading the 8018A memory, word length, and number of words from any HP-IB compatible controller. Starting and stopping of data generation is also remotely controllable.	add $650
Opt 002: 15263A Card Reader. Provides fast loading of 8018A. Data stored on punched or marked cards is loaded into the 8018A via its HP-IB interface. Requires Opt 001.	add $735
Opt 907: Front Handle Kit	add $30
Opt 908: Rack Mounting Kit	add $25
Opt 909: Rack Flange & Front Handle Combination kit	$45
Opt 910: Extra Operating and Service manual	add $20
HP-IB cables: Refer to page 28	

8018A Serial Data Generator **$3475**